```python
import numpy as np
import matplotlib.pyplot as plt

def sigmoid(z):
    return 1 / (1 + np.exp(-z))

def logistic_regression(x, y, lr, epochs):
    a = 0
    b = 0
    n = len(x)

    for _ in range(epochs):
        y_pred = sigmoid(a + b * x)
        da = (1 / n) * np.sum(y_pred - y)
        db = (1 / n) * np.sum((y_pred - y) * x)
        a -= lr * da
        b -= lr * db

    return a, b


# Sample data
x = np.array([1, 2, 3, 4, 5])
y = np.array([0, 0, 1, 1, 1])

# Train model
a, b = logistic_regression(x, y, lr=0.1, epochs=1000)

print("Intercept (a):", a)
print("Slope (b):", b)

# Prediction
x_new = 6
prob = sigmoid(a + b * x_new)
print("Predicted probability for x =", x_new, "is", prob)

# Plotting
x_vals = np.linspace(0, 7, 100)
y_vals = sigmoid(a + b * x_vals)

plt.scatter(x, y, label="Actual Data")
plt.plot(x_vals, y_vals, label="Logistic Curve")
plt.xlabel("X")
plt.ylabel("Probability")
plt.title("Logistic Regression Model")
plt.legend()
plt.grid(True)
plt.show()
```

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Intercept (a): -5.2552825294787695
Slope (b): 2.2204625990763396
Predicted probability for x = 6 is 0.9996865299021876
```